

Yeritmary Rodríguez Delgado

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Education

Carnegie Mellon University

M.S. in Automated Science: Biological Experimentation

Pittsburgh, PA

Expected May 2027

University of Puerto Rico at Cayey

B.S. in Biology

Cayey, PR

June 2025

Relevant Coursework: Machine Learning for Scientists, Automation of Scientific Research, Computational Genomics, Laboratory Methods for Automated Biology I & II

Technical Skills

Programming & Data: Python, SQL, PyTorch, NumPy, pandas, scikit-learn, matplotlib, Jupyter, Git, VS Code

Computer Vision & Image Processing: OpenCV, scikit-image, YOLOv8, image segmentation, object detection, feature extraction, quantitative image analysis, image preprocessing, ImageJ/Fiji, CellProfiler

Automation & Robotics: Opentrons Flex, Opentrons OT-2, Hamilton liquid-handling systems, Python Opentrons API, robotic liquid handling, hardware-software integration, experimental automation, Thermo Momentum

AI & Intelligent Systems: Deep learning, active learning, AI-agent development, model-guided experimental workflows

Imaging: Fluorescence microscopy, confocal microscopy, image acquisition, image-quality evaluation, intensity quantification, localization analysis, morphology analysis

Experience

Magnify Biosciences

Robotics Engineer

Pittsburgh, PA

2026 – Present

- Develop and troubleshoot Python-based robotic liquid-handling workflows on the Opentrons Flex for automated experimental protocols.
- Integrate robotic hardware, labware geometry, liquid-handling parameters, and software control to improve reliability and reproducibility of automated workflows.
- Diagnose hardware-software failures involving pipette positioning, aspiration and dispensing behavior, liquid sensing, tip handling, deck configuration, and labware calibration.
- Develop reusable automation utilities for dynamic tip selection, deck configuration, labware loading, and protocol execution.
- Develop an AI agent for laboratory automation to support automated error tracing, troubleshooting, and intelligent workflow assistance for robotic experiments.

Carnegie Mellon University

Post-baccalaureate Research Fellow, Quantitative Imaging of Connexins

Pittsburgh, PA

Jul 2024 – Jun 2025

- Applied image segmentation, fluorescence-intensity quantification, localization analysis, feature extraction, and morphological measurements across confocal microscopy datasets spanning 21 connexin isoforms.
- Quantified cell count, transfection efficiency, gap-junction plaque localization, and plaque morphology from microscopy data using Imaris and CellPose workflows.
- Designed standardized image-analysis SOPs and reproducible workflows to improve consistency, documentation, and rigor across imaging experiments.
- Presented quantitative imaging research at ASBMB 2025 and PittCell 2025.

University of Puerto Rico at Cayey

NIH U-RISE Undergraduate Research Fellow

Cayey, PR

Aug 2021 – Jul 2024

- Developed video-based behavioral analysis workflows using EthoVision to quantify movement and experimental outcomes across approximately 200 samples.
- Optimized data-acquisition parameters, experimental procedures, and analysis workflows to improve reproducibility of tracked behavioral measurements.

Relevant Projects

U-Net Image Segmentation – Automation of Scientific Research

Jan 2026 – May 2026

- Developed a U-Net convolutional neural network for image segmentation to identify and quantify regions of interest in scientific imaging data.
- Built an end-to-end image-processing pipeline for preprocessing, model training, inference, and quantitative evaluation of segmentation performance.
- Applied deep learning and computer vision methods to automate image analysis and support model-guided scientific experimentation.

Computer Vision for Pipette Accuracy – Laboratory Methods for Automated Biology

Aug 2025 – Dec 2025

- Developed a YOLOv8-based computer vision pipeline to estimate dispensed liquid volume and assess pipetting accuracy from experimental imagery.
- Applied object detection and quantitative image analysis to automate quality-control measurements for robotic liquid-handling performance.
- Integrated computer vision with laboratory automation concepts to identify inaccurate dispensing events and support automated system validation.